Advanced Dynare

Chapter VI
Full information estimation of DSGE models

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Review

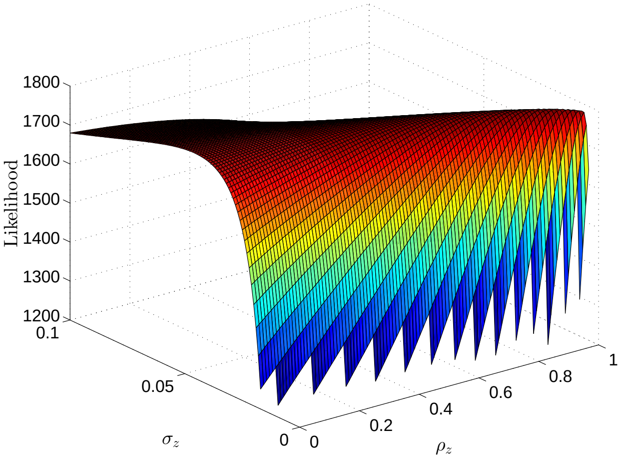
- Log-Linearized DSGE model solution takes state-state space form
- Typically, not all state variables are observed
- Solution: use Kalman Filter to deal with unobserved states and construct likelihood
- Result: likelihood $\mathcal{L}(y^T|\theta)$, where θ is the vector of deep parameters and y^T is the complete history of observables
- Estimating meant finding the parameters θ that maximized $\mathcal{L}(y^T|\theta)$

Overview

1. The problem addressed by Bayesian estimation
2. Characterizing the posterior
3. Prior distributions
4. The Metropolis-Hastings algorithm
5. Convergence and efficiency

The likelihood

- The likelihood is a high-dimensional object
- Even for simple models, it can be ill-behaved, showing hardly any curvature and exhibiting many local maxima



The likelihood

- For more complicated models, you can think of it as an egg-crate
- “Dilemma of absurd parameter estimates” (An and Schorfheide 2007): ML estimates often at odds with information from outside the model

Bayesian vs. frequentist philosophy

- In principle, there is a fundamental philosophical difference between Bayesian and frequentist econometrics
- Most macroeconomists are not Bayesian believers but pragmatists - up to the point that Bayesian and frequentist techniques are used in the same papers
- Adopting Bayesian techniques makes our lives easier
- Historically, Bayesian were the minority because computing power was not sufficient to apply their methods
- There are very good reasons to be Bayesian (see e.g. Berger and Wolpert 1988; Sims 2007)

Bayesian vs. frequentist philosophy: concept of probability

- Bayesian probability is usually associated with **degrees of belief or degrees of knowledge**
- Frequentist probability is usually based on **features of hypothetically observable systems**
  $\rightarrow$ relative frequency of an event "in the long run"
- Example: "There is a .50 probability that a fair coin will land heads."
  - Bayesian: belief evenly divided between the coin landing heads or tails
  - Frequentist: result if coin were flipped a hypothetical infinite number of times
- Frequentist analysis limited to inference about relative frequency of events in the long run
- But: long run unobservable and not in researchers' actual interest
- Bayesians can apply probability to anything that can be the subject of belief or knowledge (hypotheses, statistical parameters, entire statistical models)

Bayesian vs. frequentist philosophy: conditioning sets

- Frequentists estimate sampling distribution for estimated effect/parameter
  - distribution simulates observed effect if repeated infinite number of times, with only sampling error affecting results
  - data carry uncertainty via sampling distribution, while parameter has fixed "true" population value
    $\rightarrow$ **conditioning on parameter** $P(Y|\theta)$
  - Null Hypothesis Significance Testing (NHST) involves testing a hypothesis we do not believe in
  - p-value gives probability of effect, assuming Null is true
    $\rightarrow$ relies on hypothetically repeating experiment that never occurred

- Bayesians make direct probabilistic statements about effect/parameter, based on observed data
  - Parameter is treated as random/uncertain while data is taken as fixed
    $\rightarrow$ **conditioning on data** $P(\theta|Y)$

The posterior density

- Central object: **posterior probability/posterior density**

$$P(\theta|Y) \tag{1}$$

  of a parameter θ, conditional on having seen the data Y

- Posterior is fully-fledged density function!

- Allows statements like: the regression indicates that with 90% probability the fiscal multiplier is between 0.3 and 1.5

- NHST does not allow such statements!

- Rejecting the Null does not tell us anything about likely effect
  $\rightarrow$ only know the data is unlikely to come from a world where the Null is true
  $\rightarrow$ estimated value is then preferred

- Problem: how to obtain this posterior?

The central element: Bayes' Rule

- Consider the basic laws of probability for two events A and B:

$$p(A,B) = p(A|B)\,p(B) \tag{2}$$

$$p(A,B) = p(B|A)\,p(A) \tag{3}$$

where $p(A,B)$ is the joint probability, $p(A|B)$ is the conditional probability, and $p(B)$ the marginal probability

- Equating them results in **Bayes' Rule**

$$p(B|A) = \frac{p(A|B)\,p(B)}{p(A)} \tag{4}$$

Bayes' Rule applied

- Now apply this to a case where we want to use some data y^T to infer a parameter vector θ

$$p\left(\theta|y^T\right) = \frac{\mathcal{L}\left(y^T|\theta\right) p\left(\theta\right)}{p\left(y^T\right)} \propto \mathcal{L}\left(y^T|\theta\right) p\left(\theta\right) \tag{5}$$

- $p\left(\theta|y^T\right)$ is called the **posterior distribution**; it incorporates all we know about the parameters and incorporates data and non-data information

- $p\left(\theta\right)$ is the **prior distribution**; it is independent of the data and incorporates all non-data information on the parameters

- $p\left(y^T\right)$ is the **data density**; as it is independent of the parameters, it can be treated as a proportionality constant (except when doing model comparison)

- Equation (5) says that the **posterior is proportional to likelihood times prior**

The problem with Bayesian estimation

- Historically, Bayesians were the minority due to insufficient computing power. Why?
- $$p\left(\theta|y^T\right) = \frac{\mathcal{L}\left(y^T|\theta\right)p(\theta)}{p(y^T)} \propto \mathcal{L}\left(y^T|\theta\right)p(\theta) \tag{5}$$

  describes a **full distribution** that often is not analytically tractable
- Characterizing the posterior distribution by its mean and variance. We need to compute:

  $$\mathbb{E}\left(\theta|y^T\right) = \int \theta p\left(\theta|y^T\right)d\theta \tag{6}$$

  and

  $$\begin{aligned}\operatorname{var}\left(\theta|y^T\right) &= \mathbb{E}\left(\theta^2|y^T\right) - \left[\mathbb{E}\left(\theta|y^T\right)\right]^2 \\ &= \int \theta^2 p\left(\theta|y^T\right)d\theta - \left[\mathbb{E}\left(\theta|y^T\right)\right]^2 \end{aligned} \tag{7}$$
- We need to evaluate integrals that usually cannot be worked out analytically!

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Getting the mean

- Ingenious idea: we are typically interested in integrals of the form

$$\mathbb{E}\left(g\left(\theta\right)|y^T\right) = \int g\left(\theta\right) p\left(\theta|y^T\right) d\theta \tag{8}$$

- If we had **iid draws from the posterior**, we could simply use a law of large numbers:

$$\mathbb{E}\left(g\left(\theta\right)|y^T\right) = \int g\left(\theta\right) p\left(\theta|y^T\right) d\theta \approx \frac{1}{S}\sum_{s=1}^{S} g\left(\theta_s\right) = \hat{g}_S \tag{9}$$

- Replace integral by sum over S draws from the posterior distribution $p(\theta|y^T)$
- This is called **Monte Carlo Integration**

Numerical standard error

- How good is this estimate?
- Central limit theorem ensures that

$$\sqrt{S}\left(\hat{g}_S - \mathbb{E}\left(g\left(\theta\right)\left|y^T\right.\right)\right) \to N\left(0, \sigma_g^2\right) \tag{10}$$

- Hence,

$$\mathbb{E}\left(g\left(\theta\right)\left|y^T\right.\right) \sim N\left(\hat{g}_S, \frac{\sigma_g^2}{S}\right) \tag{11}$$

- The standard deviation

$$\sigma_g^2 = \mathrm{var}(g(\theta)|y^T) \tag{12}$$

  can be estimated by the variance in our Monte Carlo sample
- The **Numerical Standard Error (NSE)**

$$\frac{\sigma_g}{\sqrt{S}} \tag{13}$$

  is a measure of approximation error.

Some more jargon

- Even if you are not a Bayesian believer, you should be familiar with the different terminology
- Denote with $\omega = g(\theta)$ some vector of functions of the parameters θ, defined over some region Ω

Definition 1 (Credible Set)

The set $C \subseteq \Omega$ is a $100(1-\alpha)\%$ **credible set** with respect to $p(\omega|y)$ if

$$p(\omega \in C|y) = \int_C p(\omega|y)d\omega = 1 - \alpha \tag{14}$$

- We are typically interested in the smallest one, the "Bayesian equivalent to a confidence interval"

Definition 2 (Highest Posterior Density Interval)

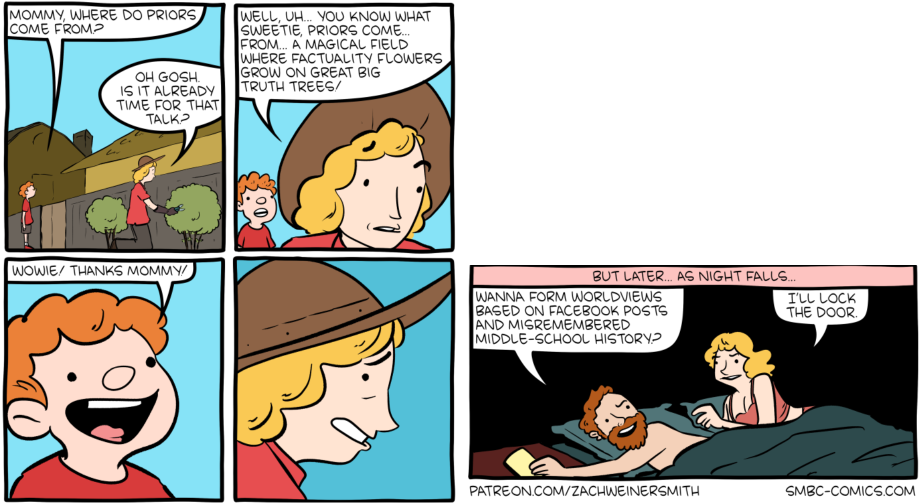
A $100(1-\alpha)\%$ **highest posterior density interval** for ω is a $100(1-\alpha)\%$ **credible interval** that has a smaller support than any other $100(1-\alpha)\%$ credible interval for ω

Problems everywhere: posterior sampling

- Monte Carlo Integration sounds nice and easy, but there's a problem: how to get draws from an intractable distribution?
- This is the big topic of **posterior sampling algorithms**
- The most important ones are
  - Importance Sampling
  - Gibbs-Sampling (S. Geman and D. Geman 1984)
  - Metropolis-Hastings algorithm (Hastings 1970; Metropolis et al. 1953)
- We will only consider the last one
- Gibbs-Sampler is a special case of the Metropolis-Hasting algorithm (see Gelman 1992)
- Gibbs sampling and the Metropolis-Hastings algorithm give rise to **correlated random draws from the posterior** and belong to the class of **Monte Carlo Markov Chain** algorithms

Overview

Where do priors come from?



Prior distributions and subjectivity

- There is a large discussion about the “subjectivity” of priors (e.g. Berger 2006)
- We will abstract from the philosophical issues arising here
- Necessarily subjective choices like e.g. the model used tend to be more important than the prior over parameters
- But be aware: issue is contentious as “subjectivity” does not square well with the “scientific method”

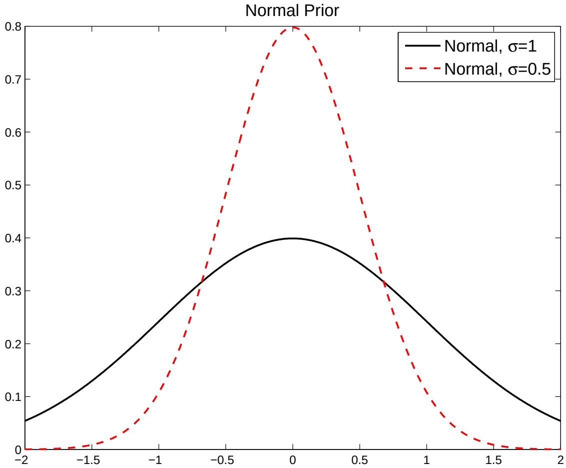
First things first: Specifying the prior

- We need to specify the prior distribution $p(\theta)$
- In general, this can be a full multivariate distribution
- In practice, people typically use **independent priors** (Andrle and Benes 2013, is a notable exception)
- Choosing sensible priors is hard, see Del Negro and Schorfheide (2008)
- What is the purpose of priors?
  1. Incorporating information extraneous to the sample
  2. Providing additional curvature to the likelihood function and straightening out cliffs
- Parameter range often narrows down prior choice
- After choosing the priors, you should do a **prior predictive**: check what the prior implies for the question you are asking (see Leeper et al. 2017)
- This way, you make sure that your estimation results are not solely driven by your prior

Endogenous priors

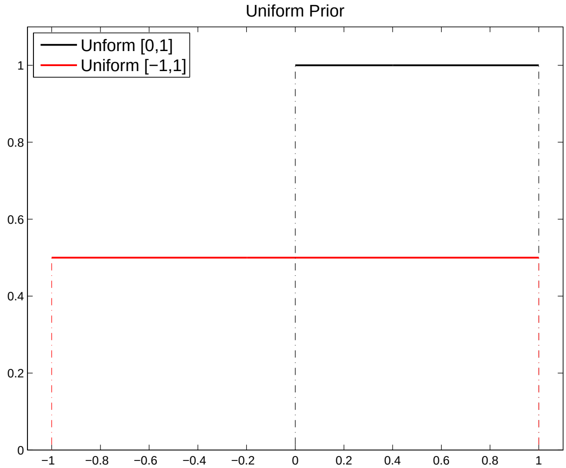
- We are doing **full information** estimation
- We are not matching moments!
- Regularly happens that estimated model implies too high variances
- Proposed solution: **endogenous priors** (Christiano et al. 2011; Del Negro and Schorfheide 2008)
- Motivated by sequential Bayesian learning
- Starting from independent initial priors, use standard deviations observed in a "pre-sample" to update those initial priors.
- Product of the initial priors and the pre-sample likelihood of the standard deviations of the observables is used as the new prior
- Triggered by the `endogenous_prior`-option of the Dynare `estimation`-command

Normal prior



- Unbounded support $(-\infty, \infty)$ and symmetric
- Typically used for e.g. feedback parameters where sign is unknown

Uniform prior



- Bounded support on $[LB, UB]$

Uniform prior

- For a variable $Y \sim U(a, b)$, the PDF is given by

$$f_U(y|a,b) = \begin{cases} \frac{1}{b-a} \text{ if } a \leqslant y \leqslant b \\ 0 \text{ otherwise} \end{cases} \tag{15}$$

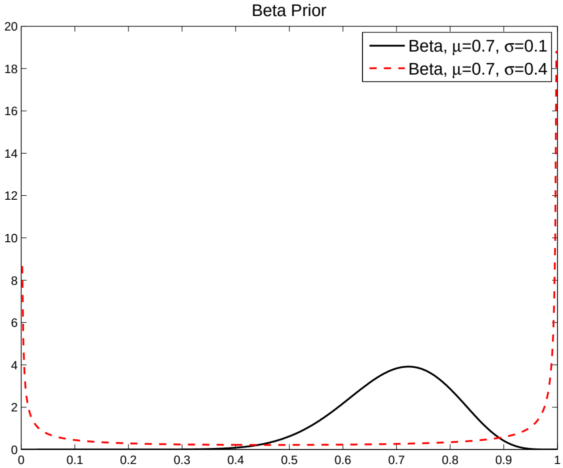
  where $-\infty < a < b < \infty$
- Moreover,

$$E(Y) = \frac{a+b}{2} \tag{16}$$

$$\mathrm{var}(Y) = \frac{(b-a)^2}{12} \tag{17}$$

- Prior is "flat", i.e. all points are equally likely, and it does not introduce curvature
- Often called **uninformative prior**
- But: it is informative in the sense that you say all parameter values in the interval are equally likely
- Beware: with bounds at infinity, prior is **improper** (cf. model comparison)

Beta prior



- Bounded support on $[0, 1]$
- Often used for autoregressive parameters, the discount factor, Calvo parameters, etc.

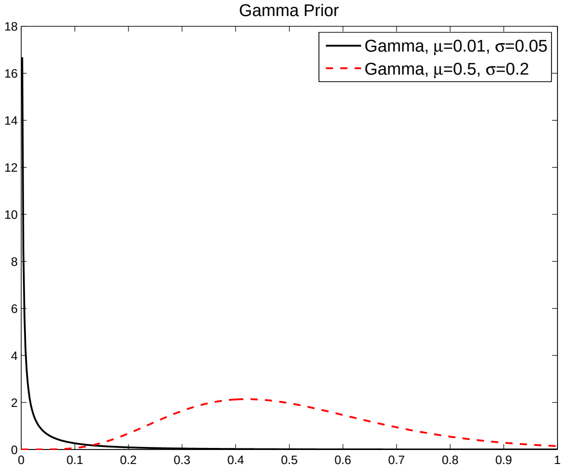
Beta prior

- Suitable transformations allow scaling it to different support like $[-1, 1]$
- Relatively flexible in allowing for different shapes, but care is required
- Due to bounded support, a high variance can result in assigning high mass to extremes in the tails

$\Rightarrow$ mode, i.e. point of highest likelihood, far away from prior mean

- Always plot the prior distribution!

Gamma prior



- Support $[0, \infty)$, i.e. 0 is included
- Typically used for variances and Taylor rule feedback parameters

Gamma prior

- For a variable $Y \sim G(\mu, \nu)$, where μ is the mean and ν the degrees of freedom, the PDF is given by

$$f_G(y|\mu,\nu) = \begin{cases} \frac{1}{\left(\frac{2\mu}{\nu}\right)^{\frac{\nu}{2}} \Gamma\left(\frac{\nu}{2}\right)} y^{\frac{\nu-2}{2}} e^{-\frac{y\nu}{2\mu}} \text{ if } 0 \leqslant y \leqslant \infty \\ 0 \text{ otherwise} \end{cases}, \tag{18}$$

  where $\Gamma()$ is the Gamma function
- Moreover,

$$E(Y) = \mu \tag{19}$$

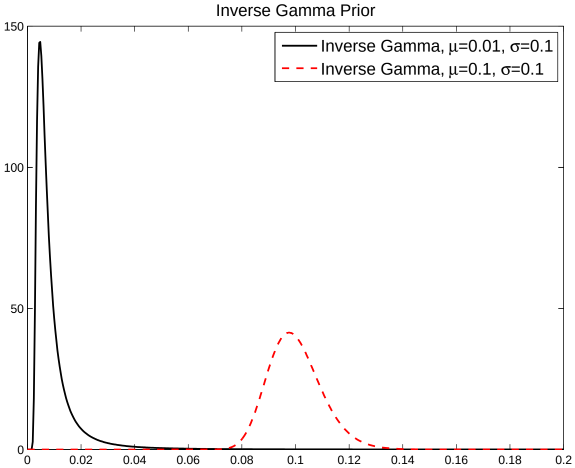
$$\text{var}(Y) = \frac{2\mu^2}{\nu} \tag{20}$$

- Beware: Matlab uses $f_G(y|a,b)$ with $a = \frac{\nu}{2}, b = \frac{2\mu}{\nu}$
- Thus, use

$$a = \frac{\mu^2}{\sigma^2} \tag{21}$$

$$b = \frac{\sigma^2}{\mu} \tag{22}$$

Inverse Gamma prior



- Support $(0, \infty)$, i.e. 0 is not included
- Typically used for variances

Inverse Gamma prior

- If y is Inverse Gamma, then $1/y$ is Gamma distributed
- The pdf is given by

$$f_{IG}(y|a,b) = \begin{cases} \frac{b^a}{\Gamma(a)} y^{a-1} e^{-by} \text{ if } 0 \leqslant y \leqslant \infty \\ 0 \text{ otherwise} \end{cases} \tag{23}$$

with

$$E(Y) = \frac{a}{b} \tag{24}$$

$$\text{var}(Y) = \frac{a}{b^2} \tag{25}$$

- Thus:

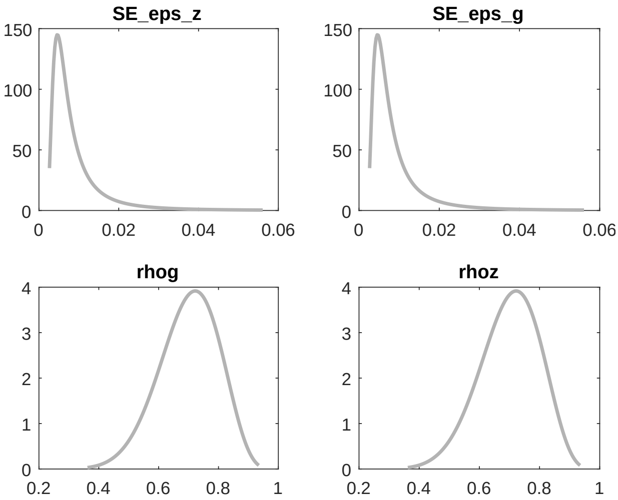
$$a = \frac{\mu^2}{\sigma^2} \tag{26}$$

$$b = \frac{\mu}{\sigma^2} \tag{27}$$

- Theoretically, a better choice for variances if you want to prevent **stochastic singularity**
- But: variance of exactly 0 is a zero probability event

Prior plots

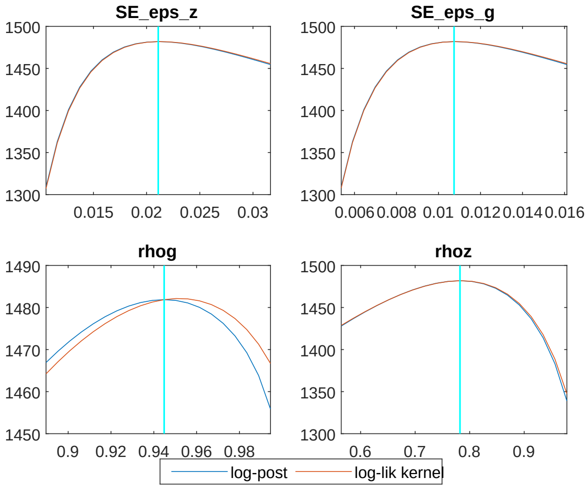
`RBC_Bayesian.mod`



- Always check whether your prior looks sensible

Mode check plots

`RBC_Bayesian.mod`



- Notice how the prior affects the posterior for `rhog`

Overview

1. The problem addressed by Bayesian estimation
2. Characterizing the posterior
3. Prior distributions
4. The Metropolis-Hastings algorithm
5. Convergence and efficiency

The problem: drawing from the posterior

- We would like to compute

$$E\left(g\left(\theta\right)|y^T\right) = \int g\left(\theta\right)p\left(\theta|y^T\right)d\theta \approx \frac{1}{S}\sum_{s=1}^{S} g\left(\theta_s\right) = \hat{g}_S \tag{28}$$

- While we can evaluate $p\left(\theta|y^T\right)$, we do not know how to draw from it
- Idea: find mechanism that
  1. draws from a stand-in proposal density
  2. evaluates both proposal density and target density
  3. reweights the draws based on the comparison of densities to obtain draws from target

Metropolis Hastings-algorithm

- Start with a vector θ_0
- Repeat for $j = 1, \ldots, N$
  - Generate candidate $\tilde{\theta}$ from proposal density $q(\theta_{j-1}, \tilde{\theta})$ and u from $\mathcal{U}(0,1)$
  - If $\tilde{\theta}$ is valid parameter draw (steady state exists, Blanchard-Kahn conditions satisfied etc.) and $u < \alpha(\theta^{j-1}, \theta^{j})$ with

$$\alpha\left(\theta_{j-1}, \tilde{\theta}\right) = \begin{cases} \min\left[\frac{p\left(\tilde{\theta}|y^T\right)q\left(\tilde{\theta}, \theta_{j-1}\right)}{p\left(\theta_{j-1}|y^T\right)q\left(\theta_{j-1}, \tilde{\theta}\right)}, 1\right] & \text{if } p\left(\theta_{j-1}\right) q\left(\theta_{j-1}, \tilde{\theta}\right) > 0 \\ 1 \text{ otherwise} \end{cases} \tag{29}$$

    set $\theta_j = \tilde{\theta}$
  - Otherwise, set $\theta_j = \theta_{j-1}$ (implies setting $p(\tilde{\theta}) = 0$ if draw invalid)
- Return the values $\{\theta_0, \ldots, \theta_N\}$
- After the chain has passed the **transient stage** and the effect of the starting values has subsided, the subsequent draws can be considered draws from the posterior

$\Rightarrow$ **burn-in** required that assures remaining chain has **converged**

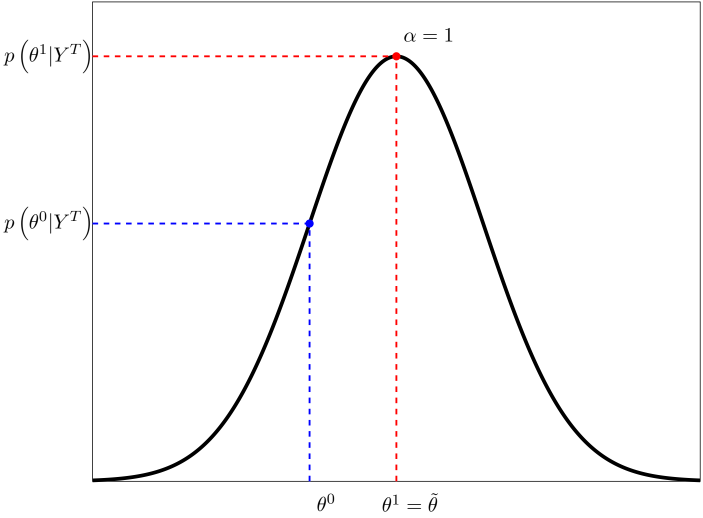
Intuition

- Note: with symmetric density, the scaling probability simplifies to

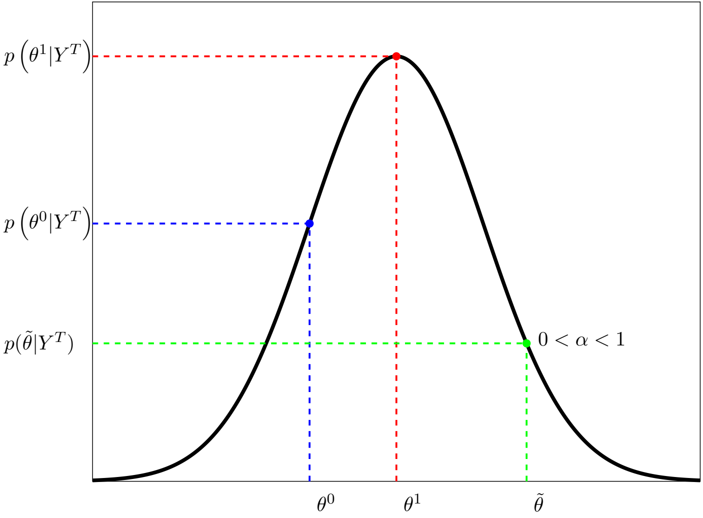
$$\alpha\left(\theta_{j-1},\tilde{\theta}\right)=\min\left[\frac{L\left(Y^T\middle|\,\tilde{\theta}\right)p\left(\tilde{\theta}\right)}{L\left(Y^T\middle|\,\theta_{j-1}\right)p\left(\theta_{j-1}\right)},1\right] \tag{30}$$

- Jump always "uphill" and "downhill" with some probability (cf. Simulated Annealing)

MH with symmetric proposal: uphill



MH with symmetric proposal: downhill



The Random-Walk Metropolis Hastings algorithm

- As long as the regularity conditions are satisfied, any proposal density q will ultimately lead to convergence to the invariant distribution
- However: speed of convergence may differ significantly
- In practice, people often use the **Random-Walk Metropolis Hastings** algorithm where

$$q\left(\theta_{j-1}, \tilde{\theta}\right) = q_{RW}\left(\theta_{j-1} - \tilde{\theta}\right) \tag{31}$$

  and q_{RW} is a multivariate density
- The candidate $\tilde{\theta}$ is thus given by the old value θ_{j-1} plus a random variable increment

$$\tilde{\theta} = \theta_{j-1} + z, z \sim q_{RW} \tag{32}$$

Choosing a proposal density

- Often, one uses
$$q_{RW} = \mathcal{N}(0, c^2\Sigma) \text{ or } q_{RW} = t_\nu(0, c^2\Sigma) \tag{33}$$
- Thus, one needs a **scaling matrix** Σ and a **scaling factor** c
- Idea: construct Gaussian approximation to posterior and use asymptotic covariance matrix as scaling matrix
- Allows efficient evaluation around mode
- Note: asymptotically, the prior plays no role and the posterior will only depend on the likelihood
- Koop (2003) only uses likelihood function, while An and Schorfheide (2007) use posterior; we side with the latter

Asymptotic normality

- Using normal approximation is justified, because with regularity conditions, posterior of θ will be **asymptotically normal** (see e.g. Crowder 1988; Kim 1998; Walker 1969)
- But convergence may be slow
- Normal approximation can often be improved by **natural re-parametrization** (Adolfson et al. 2007): bring bounded parameters to unbounded support:
  - log transformation of positive parameters
  - logit-transformation for parameters on unit interval

Technical considerations: the starting point

- Usual procedure: use numerical optimizer to find **posterior mode** of log-posterior $p(\theta|Y^T)$
- In practical applications, non-derivative based optimizers seem to perform better
- Finding the mode is hard and time-intensive; try (sequence of) different optimizers
- In some sense, finding the mode is not important as long as the regularity conditions are met (positive definite scaling matrix)
- Wherever you start, asymptotically the MCMC sampler will spend most time at the mode and get there
- MCMC is a quite inefficient optimizer (cf. Simulated Annealing)
- Thus: if you only have finite time, try to get as close as possible
- Problems of not having found the mode can often be seen as a slow drift in the parameters and posterior density

Technical considerations: the scaling matrix I

- Set the scaling matrix Σ to the **inverse Hessian** at the posterior mode:

$$\Sigma = \mathrm{var}\left(\hat{\theta}\right) = I\left(\theta\right)^{-1} = \left(-E\left[\frac{\partial^2 \log p\left(\theta|Y\right)}{\partial\theta\partial\theta'}\right]\right)^{-1} \tag{34}$$

- `csminwel/newrat` will provide an estimate of the Hessian as one of its outputs
- In practice, having fatter tails often works better: might want to use t-distribution instead of normal
- Sounds easy, but creates a lot of problems!

Technical considerations: the scaling matrix II

- Theoretical Inverse Hessian at the true mode is positive definite, but the numerical one at the conjectured mode often is not
- Literature uses various (dirty) tricks like
  - use Jordan decomposition to decompose matrix, set the eigenvalues smaller than or equal 0 to some small number, and then recompose the matrix
  - use **generalized Cholesky** (see e.g. Gill and King 2004)
  - Try different step sizes for numerical evaluation of derivatives (An and Schorfheide 2007) $\rightarrow$ `analytic_derivation`-option can also help

Non-positive definite Hessians: what to do?

- Theory (e.g. Chib and Greenberg 1995): any positive definite proposal matrix for the jumping distribution should work (only efficiency will suffer)

- Two things one can try:
  1. `mode_compute=6` runs MCMC chain and will use covariance of draws for jumping proposal density
     $\rightarrow$ almost always positive definite, but numerically problematic in case of corner solutions
  2. Use `mcmc_jumping_covariance=option`-option to start estimation with a user-defined covariance matrix:
     - `prior_variance` uses prior variances on the diagonal
     - `identity_matrix` employs identity matrix
     - provide a user-defined matrix in a `.mat`-file,

     where `mh_jscale` can be used to scale the matrices

- If resulting MCMC run performs well, recompute covariance matrix of the actual draws and use that one for new run

Choosing the scaling factor

- The scaling factor affects the behavior of the M-H Chain through:
  - **Acceptance Rate**: percentage of times a move is made
  - Region of the sample space covered by the sampler
- Consider a case where the sampler has converged and the area around the **mode** is sampled
- If the scaling is too wide, many implausible parameter vectors far away from the mode will be proposed and rejected (accept. prob. low)
- If the scaling is too small, many likely parameter vectors close to the mode will be proposed and accepted (accept. prob. high)
- Low probability regions will be undersampled $\Rightarrow$ will take the sampler a long time to traverse the support of the density
- In both cases, there tends to be a **high autocorrelation in the draws**

Choosing the scaling factor: Optimal value

- Roberts et al. (1997): if target and proposal are normal densities,
  - the optimal acceptance rate is 45% in the univariate case
  - 23% for infinitely many parameters (and already 25% for 6 parameters)
- People usually aim at a value around 25-30 percent
- A good starting point is `mh_jscale=2.38/sqrt(n)`, where n is the number of parameters estimated
- `mh_tune_jscale=double` option in Dynare to automatically tune the `mh_jscale` parameter to achieve a target

Acceptance rate too low

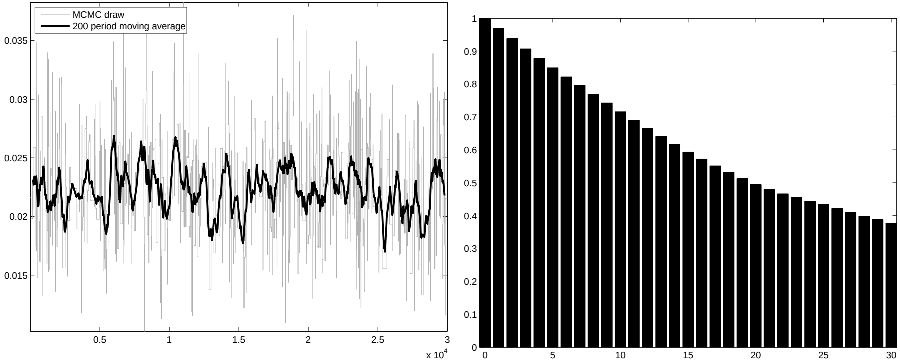


Fig. 1: Trace and Autocorrelation Plot: `eps_z`

- Acceptance Rate of 2.5%
- Bad mixing and autocorrelation function only decays slowly

Acceptance rate too high

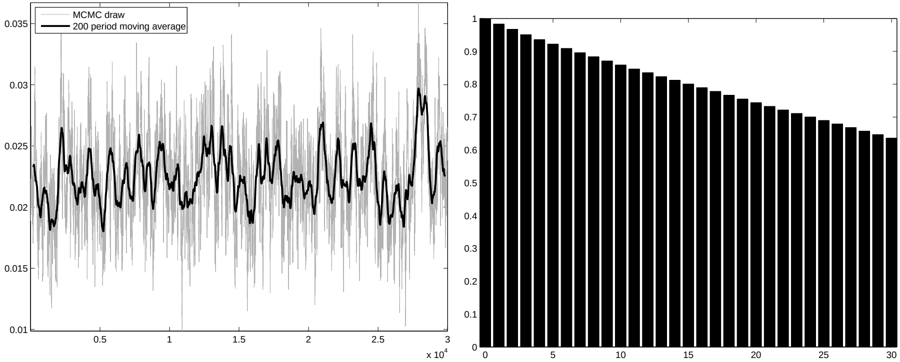


Fig. 2: Trace and Autocorrelation Plot: `eps_z`

- Acceptance Rate of 85%
- Bad mixing and autocorrelation function only decays slowly

Acceptance rate on target

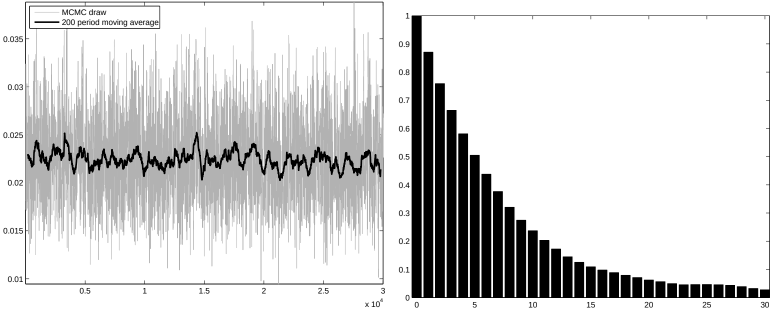


Fig. 3: Trace and Autocorrelation Plot: `eps_z`

- Acceptance Rate of 21%
- Good mixing and autocorrelation function decays relatively fast

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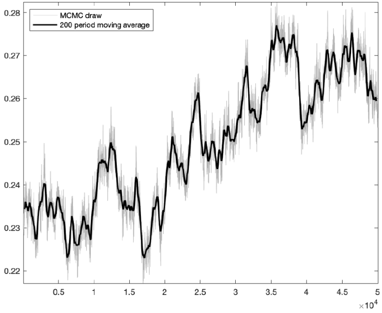
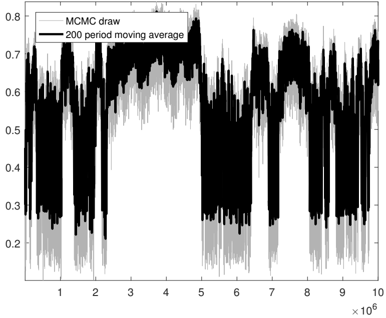
Diagnostics: trace plots





- Trace plots available via

```
generate_trace_plots(chain_number)
```

provide valuable clues to non-convergence
→ e.g. drifts to regions of higher posterior density (left) or multi-modality (right)

Efficiency

- Ideally, we want iid draws from the posterior
- MCMC only delivers correlated draws (hence: Markov Chain)
- While high autocorrelation in the draws may signify problems with the scaling, even with the "optimal" acceptance rate, there might be high autocorrelation

More powerful posterior samplers

- What if Markov Chain shows poor behavior: drift, high autocorrelation, poor mixing, large inefficiency factors?

- Try a more powerful `posterior_sampling_method`:
    1. `'slice'`: Slice sampler
    2. `'tailored_random_block_metropolis_hastings'`: Tailored Random Block Metropolis Hastings (TaRB) of Chib and Ramamurthy (2010)
    3. `'dime_mcmc'`: Differential-Independence Mixture Ensemble (DIME) sampler of Boehl (2022)
    4. `'hssmc'`: Sequential Monte-Carlo sampler of Herbst and Schorfheide (2014)
    5. `'dsmh'`: Dynamic Striated Metropolis Hastings sampler of Waggoner et al. (2016)

- Tend to be computationally more expensive but require less tweaking

```
MCMC Inefficiency factors per block
Parameter          Block 1
SE_e_a              61.868
SE_e_m              18.233
alp                133.896
bet                 40.685
gam                 28.205
mst                 93.424
rho                 38.323
psi                 50.946
del                119.450
```

Tailored Random Block Metropolis Hastings (TaRB)

- Triggered using `posterior_sampling_method='tailored_random_block_metropolis_hastings'`
- Basic idea employ multiple-block MH algorithm by grouping parameters into distinct blocks and sequentially update each block by a MH step
- Correlated parameters should be put into same block, uncorrelated sets in different blocks
- Problem: correlation structure a priori unknown
- Solution: randomly assign parameters to blocks, with the `posterior_sampler_options` name/value pair (`new_block_probability,double`) governing probability of new block
- For each MH step: run additional mode-finding step and compute (approximation) to Hessian
  $\rightarrow$ powerful for finding true mode, but computationally expensive

Monitoring convergence

- We only want to consider draws after the transition kernel has converged to the invariant distribution
- How to monitor that?
  1. Geweke (1992) convergence diagnostics: requires single MCMC
  2. Brooks and Gelman (1998) convergence diagnostics: requires at least two MCMC
  3. Raftery and Lewis (1992) convergence diagnostics: can be requested using the `raftery_lewis_diagnostics`-option

1. Geweke (1992) Convergence Diagnostics

- Idea: if we have sufficient number of draws from posterior, the first S_A draws after a burn-in of S_0 should be similar to the last S_C draws
- By leaving out S_B draws in the middle, draws in S_A and S_C should be independent
- In practice, often $S_A = 0.1S_1$, $S_C = 0.4S_1$, where S_1 is the number of draws after the burn-in
- To test similarity, test the means (or any other statistics) $g_{S_i} = E(g(\theta)|Y^T), i \in \{A, C\}$:

$$CD_{GWK} = \frac{\hat{g}_{S_A} - \hat{g}_{S_C}}{\frac{\hat{\sigma}_A}{\sqrt{S_A}} + \frac{\hat{\sigma}_C}{\sqrt{S_C}}} \tag{35}$$

- This is a two-sample t-test and thus asymptotically

$$CD_{GWK} \to N(0, 1) \tag{36}$$

- Problem: estimate of numerical standard error $\hat{\sigma}_i$ needs to take correlation in draws into account
- Use Newey and West (1987)-type estimator that tapers spectral density

Example

Geweke (1992) Convergence Tests, based on means of draws 100000 to 120000 vs 150000 to 200000.
p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper	p-val 15% Taper
SE_eps_z	0.023	0.004	0.000	0.299	0.292	0.262
SE_eps_g	0.011	0.001	0.681	0.915	0.915	0.921
rhog	0.944	0.009	0.024	0.517	0.520	0.507
rhoz	0.756	0.069	0.024	0.568	0.560	0.540

- Higher tapers correcting for serial correlation suggest convergence

2. Brooks and Gelman (1998) Convergence Diagnostics

- Idea: wherever the MC starts, it should converge to the same invariant distribution
- Start multiple chains from **over-dispersed draws** and see whether they yield similar posterior draws after burn-in
- The univariate convergence diagnostics are based on comparing **pooled** and **within** MCMC moments (ANOVA)
- Consider a statistic g with variance σ and having J chains with N draws each after discarding a burn-in
- Denote means with bars
- The **variance within a chain** is given by

$$\sigma_j^2 = \frac{1}{N-1} \sum_{n=1}^{N} (g_{nj} - \bar{g}_j)^2 \tag{37}$$

Getting the variances

- The **between sequence variance** B/N is given by

$$\frac{B}{N} = \frac{1}{J-1} \sum_{j=1}^{J} (\bar{g}_j - \bar{g})^2 \tag{38}$$

- Here, B/N is the square of the standard error of the mean and B the actual variance estimate of g
- The (average) **within sequence variance** is given by

$$W = \frac{1}{J(N-1)} \sum_{j=1}^{J} \sum_{n=1}^{N} (\bar{g}_{jn} - \bar{g}_j)^2 = \frac{1}{J} \sum_{j=1}^{J} \sigma_j^2 \tag{39}$$

- The variance σ^2 can now be estimated by a weighted average of the two:

$$\hat{\sigma}^2 = \left(1 - \frac{1}{N}\right) W + \frac{1+J}{NJ} B \tag{40}$$

- Because we use over-dispersed starting points, this is an overestimate of σ^2, but it is consistent

Potential scale reduction factor

- The statistic of interest is the **potential scale reduction factor**, i.e. the ratio between the pooled variance estimate and the within-chain variance estimate:

$$\hat{R} = \sqrt{\frac{\hat{\sigma}^2}{W}} \tag{41}$$

- If $\hat{R}$ is large, either the pooled variance estimate $\hat{\sigma}^2$ can be decreased by further simulations or the within chain variance will increase due to it not yet having made a full tour through the target distribution
- If $\hat{R}$ is close to 1, each of the J chains of N draws is close to the target distribution
- At convergence, three properties should hold
  - $\hat{R}$ should be close to 1
  - The pooled variance $\hat{\sigma}^2$ should stabilize with convergence as the chains were started from an over-dispersed distribution
  - The same should hold true for the within chain variance, which should be smaller than $\hat{\sigma}^2$
- All three conditions can be monitored graphically

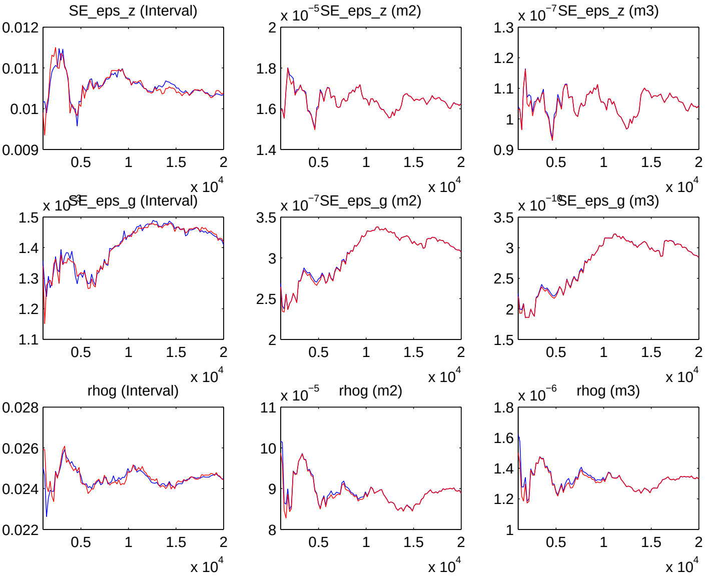
A non-parametric version

- But: previous approach assumes normality by looking at means and variances
- Alternative (also used in Dynare): take length of the $1-\alpha$ percentile for each of the chains and for the pooled draws
- Compare the interval for the pooled draws with the average from the individual chains

$$\hat{R}_{interval} = \frac{\text{length of total sequence interval}}{\text{mean length of within sequence interval}} \tag{42}$$

- $R_{interval}$ is a **potential scale reduction factor** based on percentiles
- The same convergence criteria apply

Example


SE_eps_z (Interval)
x 10^-5 SE_eps_z (m2)
x 10^-7 SE_eps_z (m3)
SE_eps_g (Interval)
x 10^-7 SE_eps_g (m2)
SE_eps_g (m3)
rhog (Interval)
x 10^-5 rhog (m2)
x 10^-6 rhog (m3)
x 10^4

3. Raftery and Lewis (1992) convergence diagnostics

- The question: how many draws do we need to make sufficiently precise statements about a quantile of a posterior function $U(\theta)$?
- Applicable to any function $U()$, but we take $U(\theta) = \theta$
- Want to estimate quantile q to within $\pm r$ with probability s
- Typically: $q = 0.025$, $r = 0.0125$, $s = 0.95$
  $\rightarrow$ want reported 95% HPDIs of U to have actual posterior probability between 92.5% and 97.5%
- To do so, we drop M draws as a burn-in and then take N draws of which we use every k^{th} one
- Central idea: construct Markov Chain out of $U(\theta_t)$ and then use Markov-Chain theory
- Helpful derivations are provided in Trairatphisan et al. (2014, SI 6) and Richard Weber's course notes

Discretization and finding thinning factor k

- First step: create binary indicator of whether a draw is below or above the cutoff value u of the q^{th} percentile

$$Z_t = \begin{cases} 1, & \text{if } U(\theta_t) \leqslant u \\ 0 & \text{otherwise} \end{cases} \tag{43}$$

- Z_t is typically not a Markov Chain, but thinned process $\left\{Z_t^{(k)}\right\}$, where we only take every k^{th} draw, will typically approximate Markov Chain for sufficiently large k
- Find k by likelihood ratio test between second- and first-order Markov model $\rightarrow$ take smallest k for which first-order model fits better
- Next step: find burn-in M

Finding the burn-in M I

- Denote the transition matrix for $\left\{Z_t^{(k)}\right\}$ with

$$P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \tag{44}$$

- The stationary distribution is given by

$$\pi = \begin{pmatrix} \pi_0 \\ \pi_1 \end{pmatrix} = \frac{1}{\alpha+\beta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \tag{45}$$

  and the l-step transition matrix by

$$P^l = \begin{pmatrix} \pi_0 & \pi_1 \\ \pi_0 & \pi_1 \end{pmatrix} + \frac{\lambda^l}{\alpha+\beta} \begin{pmatrix} \alpha & -\alpha \\ -\beta & \beta\, , \end{pmatrix} \tag{46}$$

  where $\lambda = 1-\alpha-\beta$

Finding the burn-in M II

- We want a burn-in so that after m draws, we are within ε of stationary distribution, i.e.

$$P\left(Z_m^{(k)} = i | Z_m^{(0)} = j\right) = e_j P^m e_i^T \tag{47}$$

  to be within ε of π_i for all i, j (where e_i denotes Euclidean basis vector)
- This is satisfied for

$$\lambda^m \leqslant \frac{(\alpha+\beta)\,\varepsilon}{\max(\alpha,\beta)} \Leftrightarrow m = \frac{\log\left(\frac{(\alpha+\beta)\varepsilon}{\max(\alpha,\beta)}\right)}{\log\lambda} \tag{48}$$

- Thus

$$M = mk \tag{49}$$

- Dynare uses $\varepsilon = 0.001$

Finding the required number of draws N

- The quantile estimate is given by

$$\bar{Z}_t^{(k)} \equiv \hat{P}\left(U \leqslant u\right) = \frac{1}{n}\sum_{t=1}^{n} Z_t^{(k)} \tag{50}$$

- For large n, it is approximately normally distributed with mean q and variance

$$\frac{1}{n}\frac{\left(2-\alpha-\beta\right)\alpha\beta}{\left(\alpha+\beta\right)^3} \tag{51}$$

- Requirement $P\left(q - r \leq \bar{Z}_t^{(k)} < q + r\right) = s$ implies

$$n = \frac{\left(2-\alpha-\beta\right)\alpha\beta}{\left(\alpha+\beta\right)^3}\left(\frac{\Phi^{-1}\left(\frac{1}{2}\left(s+1\right)\right)}{r}\right), \tag{52}$$

  where Φ^{-1} is the inverse standard normal CDF
- Thus

$$N = nk \tag{53}$$

Example

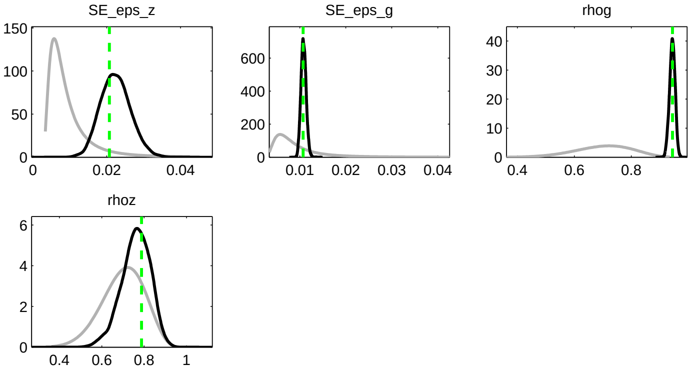
Raftery/Lewis (1992) Convergence Diagnostics, based on quantile q=0.025 with precision r=0.005 with probability s=0.950.

Variables	M (burn-in)	N (req. draws)	N+M (total draws)	k (thinning)
SE_eps_z	31	33318	33349	1
SE_eps_g	34	37136	37170	1
rhog	26	28728	28754	1
rhoz	36	39038	39074	1
Maximum	36	39038	39074	1

- Can be requested using the `raftery_lewis_diagnostics`-option

Prior vs. posterior plots

`RBC_Bayesian.mod`



- Priors (grey) and posteriors (black) differ: data seems to be informative for updating
- Exception: ρ_z where we saw the flat likelihood
- When the prior is (almost) equal to the posterior, there are two cases
  - Data is uninformative
  - Data is informative, but coincides with chosen prior

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